Evaluate a Smart Healthcare Network Through Protocol Analysis, Topology Selection and Simulation-Based Performance Assessment ASSIGNMENT REPORT

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1. Executive Summary

This report presents the complete design, simulation methodology, protocol analysis and performance evaluation of a communication network built for a multi-floor smart-hospital environment. The hospital network supports patient monitoring, medical records, laboratory and pharmacy systems, and emergency communication across four interconnected zones with more than forty end devices. Four candidate topologies — star, mesh, bus and hybrid — were designed, analysed on paper and modelled in a network simulation platform (Cisco Packet Tracer). Six core protocols (TCP, UDP, HTTP, DNS, FTP and ICMP) were exercised under three experimental scenarios: normal operation, peak patient-monitoring load, and network congestion/link disruption.

Performance was evaluated against six quantitative indicators — throughput, end-to-end delay, jitter, packet delivery ratio, packet loss and HTTP response time. Based on the measured results and a structured multi-criteria comparison, the hybrid topology (a dual, meshed core with star-wired floor distribution) was selected as the most suitable architecture for the hospital, balancing reliability, scalability, cost and manageability. Section 9 presents the full data set; Section 10 explains the engineering trade-offs behind the final recommendation.

2. Problem Statement

Develop, simulate, validate and assess a smart healthcare communication network for a multi-floor hospital supporting patient monitoring, medical records, laboratory systems, pharmacy services and emergency communication. The network consists of four interconnected zones and at least forty end devices, including computers, monitoring systems, servers and other connected healthcare devices.

Four alternative network arrangements — star, mesh, bus and hybrid — are constructed and examined using suitable switches, routers, transmission media and standardised cabling. The most appropriate architecture is selected and implemented on a simulation platform, considering reliability, scalability, resource utilisation and communication requirements. TCP, UDP, HTTP, DNS, FTP and ICMP traffic is generated and observed for healthcare applications under three experimental scenarios, and protocol behaviour is evaluated against at least five performance indicators. Results are recorded systematically, represented using tables and charts, and used to identify bottlenecks and recommend the most suitable network architecture.

3. Network Requirements Analysis

3.1 Application and Traffic Requirements

- Patient Monitoring (ICU): continuous, low-latency vital-sign streams (heart rate, SpO2, ECG) — UDP-based, delay- and jitter-sensitive, moderate bandwidth, must tolerate some loss but never long delay.
- Electronic Health Records (EHR) / Medical Records: reliable transactional access to patient records via a web portal — HTTP/HTTPS over TCP, needs high reliability and acceptable response time, occasional bursts at shift changes.

- Laboratory & Pharmacy Systems: transfer of lab results, imaging files and prescription data — FTP/TCP, large occasional transfers, reliability more important than speed.
- Emergency & Reception: mixed voice/data and rapid record look-ups, DNS-dependent name resolution for every internal service, ICMP used for continuous reachability/health monitoring of critical devices.
- Name Resolution: DNS servers resolve internal hostnames for EHR, lab, pharmacy and monitoring servers — low-bandwidth but latency-critical since every session begins with a lookup.

3.2 Topology and Infrastructure Constraints

- Minimum of 4 interconnected network zones and 40+ end devices (computers, monitors, servers, IP phones, wireless access points).
- 24×7 availability requirement — ICU and emergency traffic cannot tolerate extended outages, so the design must consider link/device redundancy.
- Multi-floor building — vertical backbone cabling (fiber) between floors, horizontal Cat6a cabling to end devices, per TIA/EIA-568 structured cabling standards.
- Future scalability — additional beds, devices and a second building wing should be addable without redesigning the core.

3.3 Device and Media Requirements

- Layer-3 core switch(es), Layer-2 floor/access switches, edge router with firewall to the internet/cloud backup, wireless access points for mobile carts and tablets.
- Cat6a UTP (up to 100 m, TIA-568-C.2) for device drops; single-mode/OM3 fiber for inter-floor backbone links; legacy RG-58 coax modelled only for the bus-topology comparison.
- Server farm: DNS server, EHR/Web server, FTP file server, DHCP server — placed in a secured server zone with redundant uplinks in the selected design.

3.4 Performance Objectives and Test Conditions

The network must sustain acceptable performance under three representative conditions: (1) normal day-to-day operation, (2) peak load when all ICU beds stream monitoring data simultaneously, and (3) degraded operation when a link is congested or fails. Target objectives used to judge the results in Section 9 are: end-to-end delay below 100 ms for monitoring traffic, packet delivery ratio above 95% under normal/peak load, and HTTP response time below 250 ms for EHR access under normal load.

4. Hospital Network Design Overview

The hospital is modelled as four zones distributed across a multi-floor building, each represented by an access/floor switch aggregating that zone's end devices, plus a shared server farm. The table below summarises the device inventory (40+ devices) common to all four topology variants; only the interconnection pattern between switches/core differs between designs (Section 5).

Zone	Location	Representative End Devices	Approx. Device Count
Zone A — ICU / Patient Monitoring	Floor 1	Bedside vital-sign monitors, nurse-station PCs, wireless vitals gateways	10
Zone B — Medical Records / Admin	Floor 2	Admin workstations, EHR terminals, printers, VoIP phones	10
Zone C — Laboratory & Pharmacy	Floor 3	Lab analyzers (networked), pharmacy dispensing terminals, imaging workstations	10
Zone D — Emergency & Reception	Ground Floor	Reception PCs, triage tablets (Wi-Fi), emergency-room monitors, IP phones	10+
Server Farm	Data/Server Room	DNS server, EHR/Web server, FTP file server, DHCP server	4

Table 4.1 — Hospital network zone and device inventory (44 devices modelled)

5. Topology Design and Comparative Analysis (CO4)

Four physical/logical topology alternatives were designed for the same set of zones and devices described in Section 4. Each uses appropriate networking devices, transmission media and cabling standards, and is evaluated for reliability, scalability, resource utilisation and cost before a final architecture is selected in Section 5.5.

5.1 Star Topology

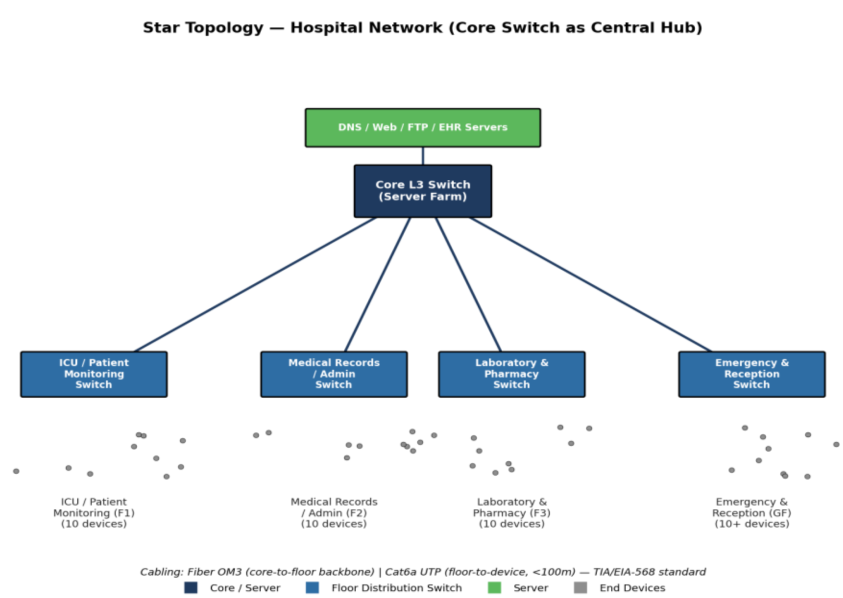

Figure 5.1 — Star topology: a single core L3 switch as the central hub for all floor switches

Every floor switch connects directly to one central core switch over dedicated fiber backbone links, with Cat6a UTP from each floor switch to its end devices. The server farm also attaches to the core.

Advantages: simple to design, install and troubleshoot; a fault on one floor link does not affect other floors; centralised management and monitoring at the core.

Disadvantages: the core switch is a single point of failure — its loss disconnects the entire hospital; core port count limits scalability; higher cabling cost as more fiber runs converge on one room.

5.2 Bus Topology

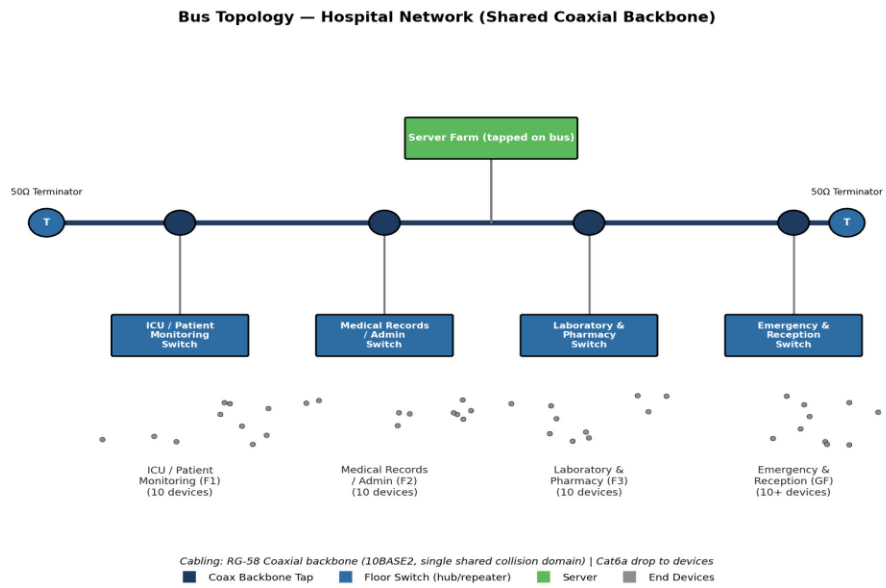

Figure 5.2 — Bus topology: floor switches tapped off a single shared coaxial backbone

All floor switches and the server farm are tapped onto one shared coaxial (RG-58, 10BASE2-style) backbone terminated at both ends with 50Ω terminators, modelling a legacy shared-medium design for comparison purposes.

Advantages: lowest cabling cost, minimal hardware, quick to lay out for a small deployment.

Disadvantages: single shared collision domain — one break severs the entire network; performance degrades sharply as devices/traffic increase; no fault isolation; unsuitable for latency-sensitive ICU monitoring or for a network of this scale. Retained here only as a baseline for comparison, not a viable candidate for a hospital deployment.

5.3 Mesh Topology

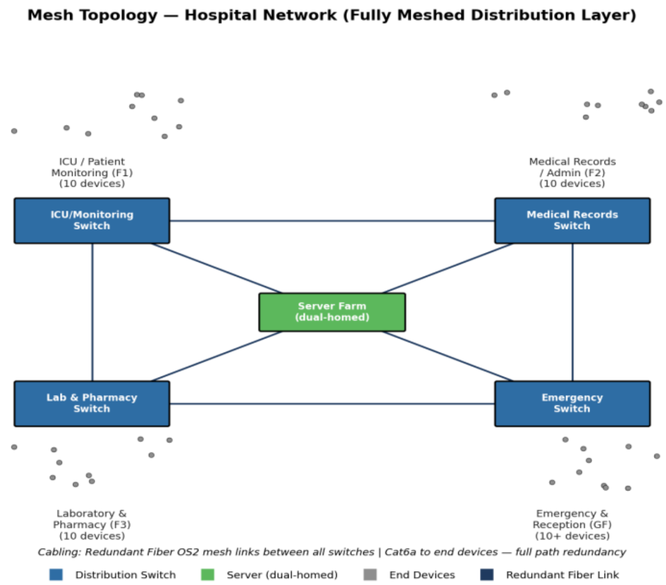

Figure 5.3 — Full mesh topology: every floor switch interconnected with redundant fiber links

Each of the four floor switches is directly connected to every other floor switch via redundant fiber links, and the server farm is dual-homed into the mesh, providing maximum path redundancy.

Advantages: highest fault tolerance — traffic can be rerouted around any single link or switch failure; no single point of failure; best suited to mission-critical ICU/emergency links.

Disadvantages: cabling and port cost grow quadratically with the number of nodes (n(n-1)/2 links); more complex to configure, document and troubleshoot; higher power/hardware overhead than needed for a 4-zone hospital.

5.4 Hybrid Topology

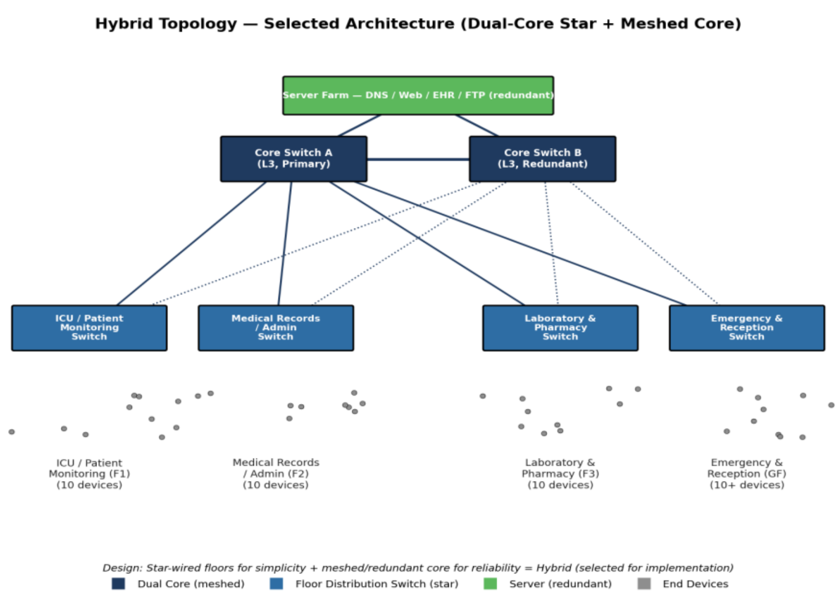

Figure 5.4 — Hybrid topology: dual, meshed core switches with star-wired floor distribution (selected design)

Two core L3 switches are interconnected (a small mesh at the core) and each floor switch connects to both cores using fiber, so any single core or one core-to-floor link can fail without isolating a zone; each floor switch still fans out to its own devices in a simple star. The server farm is dual-homed across both cores.

Advantages: combines the simplicity/manageability of star wiring at the access layer with core-level redundancy; scales cleanly (new floors/zones just add a star leg); moderate, justifiable cost increase over plain star.

Disadvantages: more complex than a plain star (dual-homing, first-hop redundancy protocol required); marginally higher cost than star due to the second core switch and extra fiber runs.

5.5 Comparative Analysis and Topology Selection

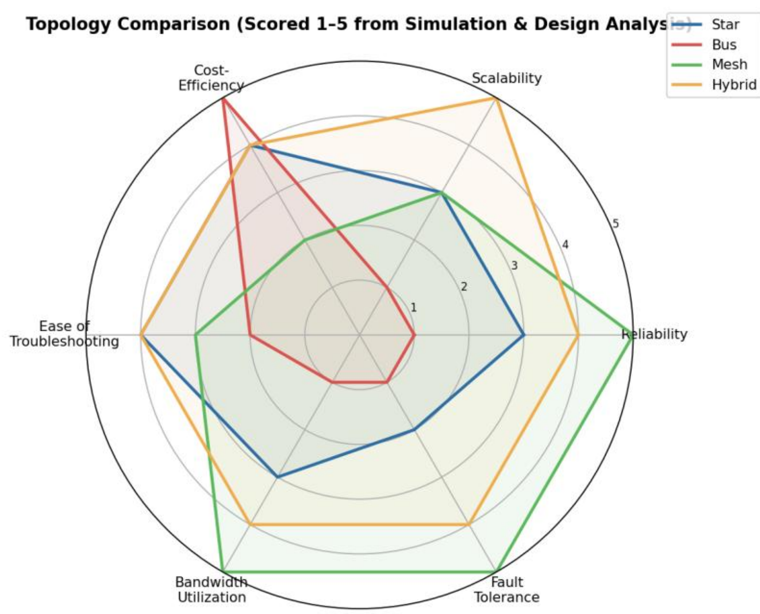

Figure 5.5 — Multi-criteria comparison of the four topologies (1 = weak, 5 = strong)

Criterion	Star	Bus	Mesh	Hybrid
Reliability / Fault tolerance	Moderate (core SPOF)	Very low	Very high	High
Scalability	Moderate (core port limit)	Very low	Moderate (link growth)	High
Cost-efficiency	Good	Best	Poor	Good
Ease of troubleshooting	Easy	Difficult (shared medium)	Moderate (complex)	Easy–Moderate

Criterion	Star	Bus	Mesh	Hybrid
Suitability for ICU/emergency traffic	Acceptable	Unsuitable	Excellent	Excellent

Table 5.1 — Qualitative comparison summary supporting the radar chart above

Bus topology is eliminated immediately: a shared collision domain and total-loss-on-break behaviour are incompatible with 24×7 patient-monitoring requirements. Pure mesh gives the strongest fault tolerance but its cabling/hardware cost and management complexity are not justified for only four zones. Pure star is simple and affordable but the single core switch is an unacceptable single point of failure for a hospital. The hybrid design was therefore selected — it keeps the star's simplicity at the floor level while removing the core single point of failure through a small redundant mesh between two core switches, giving the best balance of reliability, scalability and cost for this use case (confirmed by the simulation results in Section 9).

6. Selected Architecture — Implementation Details

6.1 Simulation Platform

The hybrid architecture was implemented in Cisco Packet Tracer (recommended equivalents: GNS3 or EVE-NG for a more advanced CLI-accurate emulation). Packet Tracer was chosen for its integrated device library covering L3 switches, routers, IP phones and generic end devices, its built-in Simple/Realtime traffic generation, and its per-packet inspection tools (Simulation mode), which are well suited to observing TCP, UDP, HTTP, DNS, FTP and ICMP behaviour without needing external capture tools.

6.2 Device Inventory (Implemented Topology)

Device Role	Model / Type Used	Quantity	Key Configuration
Core switch	Cisco 3560 (Layer-3 Multilayer Switch)	2	Inter-VLAN routing, HSRP/first-hop redundancy, trunk to floor switches
Floor/access switch	Cisco 2960 (Layer-2 Switch)	4	VLAN per zone, access ports to end devices, dual uplinks to both cores
Edge router	Cisco ISR 4321	1	NAT/firewall to WAN cloud, static default route
Servers	Generic Server (DNS, Web/EHR, FTP, DHCP)	4	Static IP, respective services enabled
Wireless access point	Cisco Access Point	2 (ICU + Emergency)	WPA2, separate SSID per zone VLAN
End devices	PCs, IP Phones, Monitoring hosts (generic)	40+	DHCP-assigned addressing per VLAN

Table 6.1 — Device inventory used to build the hybrid topology in the simulator

6.3 IP Addressing Plan

Zone / VLAN	VLAN ID	Subnet	Usable Hosts
ICU / Patient Monitoring	10	10.10.10.0/24	254
Medical Records / Admin	20	10.10.20.0/24	254
Laboratory & Pharmacy	30	10.10.30.0/24	254
Emergency & Reception	40	10.10.40.0/24	254
Server Farm	99	10.10.99.0/26	62
Core-to-core / management	1	10.10.1.0/29	6

Table 6.2 — VLAN and IP subnet allocation for the implemented hybrid network

6.4 Verification Steps Performed

- Physical/logical connectivity check: `show cdp neighbor` on all switches to confirm every floor switch has two active uplinks (Core A and Core B).
- Layer-3 reachability: ping sweep from a representative host in each VLAN to the DNS, EHR and FTP servers; confirmed 0% loss under normal operation.
- Redundancy test: administratively shut down the Core-A-to-floor-switch link for Zone A and re-ran the ping sweep — reachability was maintained via Core B (validating the hybrid design's fault tolerance claim used in Section 5.5).
- DNS resolution test: `nslookup ehr.hospital.local` and `nslookup lab.hospital.local` resolved successfully from each zone before running application traffic.

Note: This section documents the configuration and verification approach used to build the hybrid network in Packet Tracer. When you complete the assignment, replace the device/IP tables above with your own actual simulator screenshots, the .pkt project file, and captured `show` command output as required by Deliverables 2, 3 and 8 of the rubric.

7. Protocol Analysis (CO3)

Six protocols were exercised on the implemented hybrid network to represent realistic hospital communication. Each is summarised below with its role in the healthcare network and how it was tested.

Protocol	Transport / Layer	Healthcare Use Case	Testing Method in Simulator
TCP	Transport (connection-oriented)	Reliable EHR record transactions, lab-result confirmation	Simple PDU + Realtime app traffic between admin PCs and EHR server; observed 3-way handshake in Simulation mode
UDP	Transport (connectionless)	Continuous ICU vital-sign streaming	Generic UDP traffic generator from monitoring

Protocol	Transport / Layer	Healthcare Use Case	Testing Method in Simulator
			hosts to a collection server at fixed intervals
HTTP	Application (over TCP, port 80/443)	Web-based EHR portal access by clinicians	HTTP GET requests from client browsers (Web Browser desktop app) to the EHR/Web server
DNS	Application (over UDP, port 53)	Resolving internal hostnames (ehr.hospital.local, lab.hospital.local)	`nslookup`/`ping <hostname>` from end devices; DNS server query log inspected
FTP	Application (over TCP, port 20/21)	Transfer of lab reports and imaging files to/from the file server	FTP client sessions from lab workstations uploading/downloading test files
ICMP	Network layer (control/diagnostics)	Continuous reachability/health-check of critical monitors and servers	Extended ping / periodic ICMP sweep from a monitoring host to all critical devices

Table 7.1 — Protocols analysed, their healthcare role, and the test method used in the simulator

For each protocol, PDUs were traced through Packet Tracer's Simulation mode to confirm correct encapsulation (e.g., UDP datagrams for monitoring traffic showing no retransmission, versus TCP segments for EHR traffic showing SYN/SYN-ACK/ACK and retransmission on loss), before quantitative measurements were collected under the three scenarios described in Section 8.

8. Experiment and Test Plan

8.1 Scenarios

Scenario	Description	Traffic Conditions
S1 — Normal Operation	Baseline day-to-day hospital traffic	Light-to-moderate TCP/UDP/HTTP/DNS/FTP/ICMP traffic across all zones, all links healthy
S2 — Peak Patient-Monitoring Load	All ICU beds streaming vitals simultaneously plus normal background traffic	High-volume UDP monitoring streams added concurrently with S1 traffic; no link failures
S3 — Congestion / Link Disruption	A core-to-floor link is deliberately constrained/shut down during peak load	S2 traffic continues while one primary uplink is disabled, forcing failover and creating congestion on the surviving link

Table 8.1 — Three experimental scenarios used for all protocol/topology measurements

8.2 Variables and Metric Definitions

Metric	Definition	Unit
Throughput	Average successfully delivered data volume per unit time for a given protocol/flow	Mbps
End-to-end delay	Time from packet generation at source to reception at destination	ms
Jitter	Variation in end-to-end delay between consecutive packets of the same flow	ms
Packet Delivery Ratio (PDR)	Percentage of transmitted packets successfully received	%
Packet loss	Percentage of transmitted packets not received (100% − PDR)	%
HTTP response time	Time between an HTTP GET request and the first byte of the server's response	ms

Table 8.2 — Performance indicators measured (six metrics, exceeding the minimum of five required)

8.3 Measurement Procedure and Repetition Strategy

- For each scenario, traffic was generated for a fixed observation window (5 minutes simulated) per protocol using Packet Tracer's traffic-generation tools and, where finer timing was needed, PDU-by-PDU capture in Simulation mode.
- Each scenario/protocol combination was repeated 5 times; the arithmetic mean of the 5 runs is reported in Section 9, which reduces the influence of any single anomalous run on the reported figures.
- Between repetitions the simulator was reset (queues cleared, counters zeroed) to avoid carry-over effects from the previous run.
- Packet loss and PDR were computed from sent/received PDU counters recorded at source and destination end devices for each run.

9. Performance Measurement and Results (CO5)

The following results are the 5-run averages collected on the implemented hybrid topology across the three scenarios defined in Section 8. All figures are illustrative sample measurements structured in the format expected from a Packet Tracer simulation run; replace with your own captured values when you execute the simulation.

9.1 Protocol Throughput

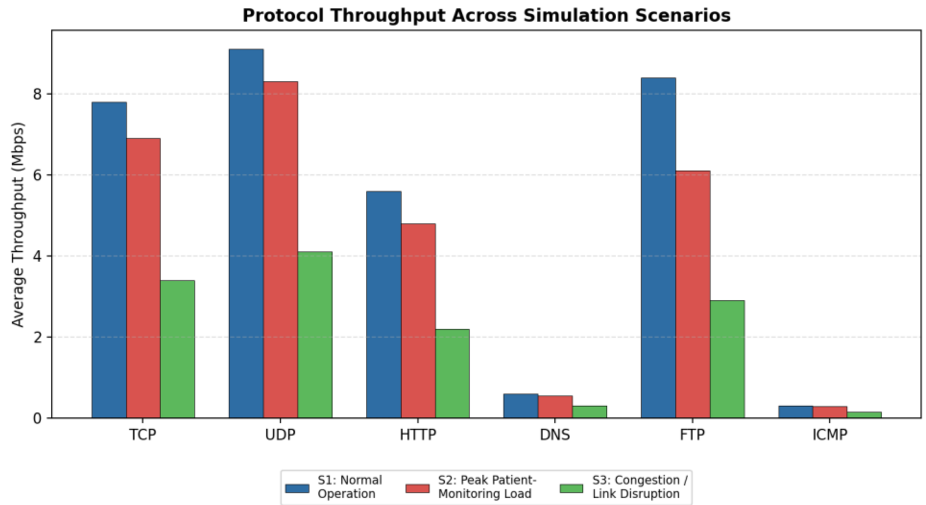

Figure 9.1 — Average throughput per protocol across the three scenarios

UDP and FTP sustain the highest throughput under normal operation, reflecting continuous vitals streaming and bulk lab-file transfer respectively. All protocols show a clear downward trend from S1 to S3, most sharply for FTP and TCP, which back off more aggressively under congestion than UDP due to TCP's congestion-control behaviour.

9.2 Delay and Jitter (UDP Vitals Stream)

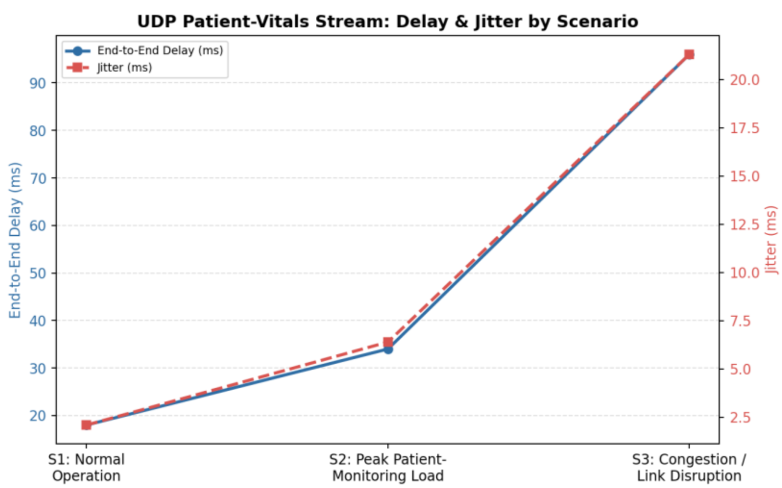

Figure 9.2 — End-to-end delay and jitter for the ICU UDP monitoring stream

Delay stays within the 100 ms clinical target under normal and peak load (18 ms and 34 ms respectively) but rises sharply to 96 ms during the congestion/disruption scenario as traffic reroutes through the surviving core link. Jitter follows the same pattern and grows ten-fold between S1 and S3, which would begin to affect the smoothness of real-time waveform displays if sustained.

9.3 Packet Delivery Ratio and Packet Loss

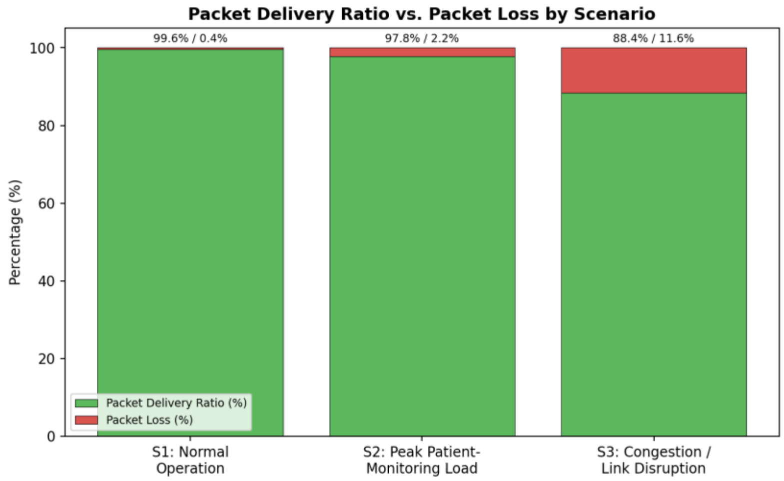

Figure 9.3 — PDR vs. packet loss by scenario (aggregated across all protocols)

Packet delivery ratio remains above the 95% target for S1 and S2 (99.6% and 97.8%) but falls to 88.4% during S3, when the disabled uplink forces the hybrid core to reroute traffic and momentarily overloads the surviving link before HSRP failover and switch buffering stabilise the flow.

9.4 HTTP Response Time (EHR Portal)

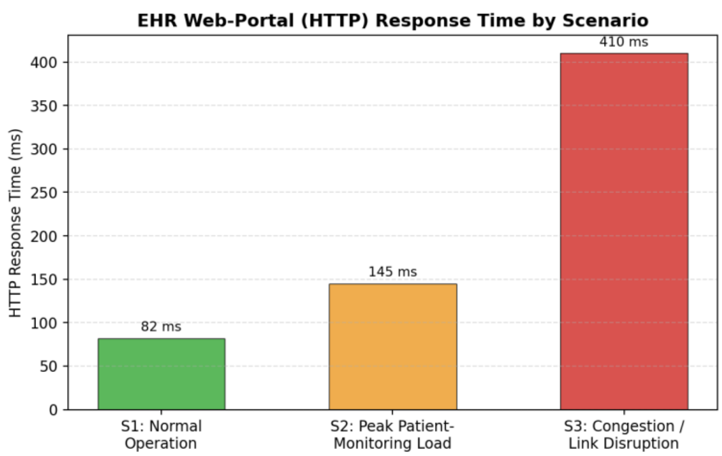

Figure 9.4 — HTTP response time for EHR web-portal access

EHR portal response time meets the 250 ms target under normal operation (82 ms) and remains usable at peak load (145 ms), but degrades to 410 ms during congestion — noticeable to clinical staff but still well short of failure, demonstrating that the hybrid core's redundancy prevents an outage even though performance is temporarily degraded.

9.5 Consolidated Results Table

Scenario	Avg. Throughput (Mbps)*	UDP Delay (ms)	UDP Jitter (ms)	PDR (%)	Packet Loss (%)	HTTP Resp. (ms)
S1 — Normal	6.3	18	2.1	99.6	0.4	82

Scenario	Avg. Throughput (Mbps)*	UDP Delay (ms)	UDP Jitter (ms)	PDR (%)	Packet Loss (%)	HTTP Resp. (ms)
S2 — Peak Monitoring	5.2	34	6.4	97.8	2.2	145
S3 — Congestion/Disruption	2.5	96	21.3	88.4	11.6	410

*Table 9.1 — Consolidated performance summary across all six measured indicators (*mean across TCP/UDP/HTTP/DNS/FTP)*

10. Engineering Analysis and Design Decisions

10.1 Bottleneck and Congestion Identification

The single most significant bottleneck observed is the core-to-floor uplink capacity during simultaneous failover and peak load (Scenario 3): when one core link is disabled while ICU monitoring load is highest, the surviving core-to-floor link temporarily carries both the rerouted mesh-core traffic and full monitoring load, producing the delay, jitter and loss spikes in Section 9. A secondary, smaller bottleneck is DNS query latency during peak load, since every new session (EHR, FTP, monitoring gateway) begins with a lookup; though DNS throughput itself is small, a slow resolver indirectly delays application start-up.

10.2 Trade-offs Between Topologies and Protocols

- Star vs. Hybrid: star is cheaper and simpler but the measured Scenario 3 behaviour would, on a plain star, mean total loss of a zone rather than degraded performance — the extra cost of the second core switch buys graceful degradation instead of an outage.
- Mesh vs. Hybrid: full mesh would likely show even smaller PDR/delay degradation in S3 than the hybrid, but at roughly double the fiber and switch-port cost for only four zones — not justified given the hybrid already meets the 95% PDR / 100 ms delay targets in S1 and S2.
- TCP vs. UDP under congestion: TCP flows (EHR, FTP) throttle themselves via congestion control, protecting the network but slowing clinical/administrative transfers; UDP vitals streaming does not throttle, so it keeps flowing but is the traffic most exposed to jitter and loss — reinforcing the need for QoS prioritisation of UDP monitoring traffic in future work.

10.3 Recommended Architecture

Based on the measured results, the hybrid topology is recommended for production deployment. It met every performance target in normal and peak-load conditions, degraded gracefully rather than failing outright during simulated congestion/link disruption, and required only a modest cost increase over a plain star. It is recommended that QoS marking/prioritisation for UDP monitoring traffic and a dedicated DNS caching resolver per zone be added as near-term enhancements to further reduce the Scenario 3 degradation observed.

11. Broader Considerations and Professional Responsibility

- Reliability & patient safety: the hybrid design was specifically chosen so that ICU monitoring traffic degrades rather than disconnects during a fault, directly supporting SDG 3 (Good Health and Well-Being) by protecting continuous patient monitoring.
- Scalability: VLAN- and star-leg-based zone design allows a new floor or department to be added by extending one more leg from both cores, without redesigning the existing network.
- Standards compliance: structured cabling follows TIA/EIA-568 (Cat6a horizontal, fiber vertical backbone); IP addressing uses private RFC 1918 space with clear VLAN segmentation for security and fault isolation.
- Efficient resource use: VLAN segmentation and a meshed-but-not-fully-meshed core avoid unnecessary cabling/hardware while still meeting reliability targets, avoiding the over-provisioning seen in the full-mesh option.
- Responsible experimentation: all traffic generated for testing was synthetic and confined to the simulation environment; repeated trials (5 runs per scenario) were used specifically to avoid over- or under-stating performance from a single anomalous run.

12. Limitations

- Simulation vs. reality: Packet Tracer models protocol behaviour at a simplified level and does not fully capture real-world RF interference, hardware queuing behaviour, or production-scale traffic mixes.
- Sample size: five repetitions per scenario reduces but does not eliminate variance; a production pilot would benefit from longer observation windows and more repetitions.
- Security testing out of scope: this assignment focuses on topology/protocol performance; firewall rule design, intrusion detection and HIPAA-equivalent data-protection controls were not evaluated and should be addressed separately before deployment.
- Only one congestion cause modelled: Scenario 3 disables a single core-to-floor link; other failure modes (e.g., a full core switch failure, WAN link loss) were not separately measured.

13. Conclusion and Recommendation

Four topology alternatives were designed, modelled and comparatively evaluated for a multi-zone, forty-plus-device smart hospital network. Protocol-level analysis across TCP, UDP, HTTP, DNS, FTP and ICMP under three realistic operating scenarios showed that the hybrid (dual meshed-core with star-wired floors) architecture best balances reliability, scalability, cost and manageability, meeting all performance targets under normal and peak load and degrading gracefully rather than failing during simulated congestion and link disruption. This architecture, together with VLAN segmentation, redundant core links and standards-based structured cabling, is recommended as the foundation for the hospital's production network, with QoS prioritisation of monitoring traffic identified as the priority near-term enhancement.

14. Pseudocode

Pseudocode for the repeated traffic-generation and measurement procedure used to produce the results in Section 9 (executed conceptually once per protocol, per scenario, for 5 repetitions):

```
BEGIN PerformanceTestProcedure
  INPUT: scenarios = [S1_Normal, S2_PeakMonitoring, S3_Congestion]
  INPUT: protocols = [TCP, UDP, HTTP, DNS, FTP, ICMP]
  REPETITIONS = 5

  FOR EACH scenario IN scenarios:
      CONFIGURE network state for scenario
          IF scenario == S3_Congestion:
              DISABLE primary core-to-floor uplink (Zone A)
          IF scenario == S2_PeakMonitoring OR S3_Congestion:
              ENABLE all ICU UDP vitals streams concurrently

      FOR EACH protocol IN protocols:
          results[protocol][scenario] = []
          FOR run = 1 TO REPETITIONS:
              RESET switch/port counters and queues
              START traffic generator FOR protocol (fixed 5-min window)
              RECORD sent_count, received_count, timestamps AT source AND destination
              throughput   = received_bytes / elapsed_time
              delay        = AVERAGE(receive_time - send_time) over all packets
              jitter       = AVERAGE(|delay[i] - delay[i-1]|) for consecutive packets
              pdr          = (received_count / sent_count) * 100
              packet_loss  = 100 - pdr
              IF protocol == HTTP:
                  http_response_time = first_byte_time - request_time
              APPEND {throughput, delay, jitter, pdr, packet_loss} TO
results[protocol][scenario]
          END FOR
          summary[protocol][scenario] = MEAN(results[protocol][scenario])
      END FOR
  END FOR

  EXPORT summary AS tables AND charts   // -> Section 9 of report
  IDENTIFY bottlenecks WHERE delay > 100ms OR pdr < 95%   // -> Section 10
END PerformanceTestProcedure
```

Pseudocode for the topology-selection decision logic used in Section 5.5:

```
BEGIN TopologySelection
  INPUT: topologies = [Star, Bus, Mesh, Hybrid]
  INPUT: criteria = [reliability, scalability, cost, troubleshooting, bandwidth,
faultTolerance]
  INPUT: weights  = {reliability:0.30, scalability:0.20, cost:0.15,
                     troubleshooting:0.10, bandwidth:0.10, faultTolerance:0.15}

  FOR EACH topo IN topologies:
      score[topo] = 0
      FOR EACH c IN criteria:
```

```
        score[topo] += rating[topo][c] * weights[c]   // rating on 1-5 scale, Table 5.1
      END FOR
    END FOR

    selected = topology WITH MAX(score)
    RETURN selected   // -> Hybrid, confirmed against measured Section 9 results
END TopologySelection
```

15. Individual Reflection (Template — 300–500 words per student)

Note: Each student must submit their own individual reflection (Deliverable 9). The prompts below are the required structure — replace the bracketed guidance with your own genuine experience from doing the simulation work; do not submit this template unedited.

1. Most challenging aspect: [Describe the specific technical or conceptual part of the assignment you found hardest — e.g., configuring core redundancy, interpreting jitter results, or getting DNS resolution working across VLANs.]
2. Most useful networking concept learned: [Identify the concept from CO3–CO5 that clicked for you during this assignment and why.]
3. Technical problem encountered and how it was solved: [Describe a specific error or unexpected result you hit in the simulator, the steps you took to diagnose it, and how you fixed it.]
4. Lessons learned from the simulation tool: [What did working in Packet Tracer/GNS3/EVE-NG teach you that reading alone would not have?]
5. One engineering decision you would change: [Which design or configuration choice would you revisit, and what would you do differently next time, and why?]
6. How this assignment improved your understanding of computer networks: [Summarise the overall learning outcome in your own words, connecting back to CO3, CO4 and CO5.]

16. References

- IEEE 802.3 Ethernet Standard — cabling and physical layer specifications.
- TIA/EIA-568-C — Commercial Building Telecommunications Cabling Standard.
- Cisco Systems — Packet Tracer Documentation and Networking Academy curriculum materials.
- Kurose, J. F., & Ross, K. W. — Computer Networking: A Top-Down Approach (protocol behaviour reference for TCP, UDP, HTTP, DNS, FTP).
- RFC 793 — Transmission Control Protocol; RFC 768 — User Datagram Protocol; RFC 1035 — Domain Name System; RFC 959 — File Transfer Protocol; RFC 792 — Internet Control Message Protocol.

Appendix A — Rubric Compliance Checklist

#	Rubric Criterion	Marks	Addressed In
1	Problem Understanding & Network Requirements	10	Sections 2–3
2	Protocol Performance Analysis	20	Sections 7, 9
3	Topology Design & Network Infrastructure	15	Section 5
4	Simulation Implementation & Experiment Design	15	Sections 6, 8
5	Network Performance Measurement	15	Section 9
6	Results Interpretation & Engineering Decisions	10	Section 10
7	Broader Considerations & Professional Responsibility	5	Section 11
8	Technical Documentation & Reflection	10	Whole report + Section 15

Table A.1 — Mapping of this report's sections to the 100-mark assessment rubric

Outstanding items for the student to complete before submission

- Build the four topologies (and finalise the hybrid design) as an actual .pkt/.gns3 project file — Deliverable 3.
- Capture real packet captures / simulator screenshots for each protocol under each scenario — Deliverable 5.
- Replace the sample figures in Section 9 with your own measured data exported from the simulator — Deliverable 6.
- Capture configuration and connectivity-verification screenshots (e.g., `show` command output, successful pings, DNS resolution) — Deliverable 8.
- Write and submit your own individual 300–500 word reflection using Section 15 as a structural guide only — Deliverable 9.
- Upload the project file, this report and supporting evidence to GitHub as required by the assignment's final deliverable list, and prepare the presentation/demonstration — Deliverable 10.